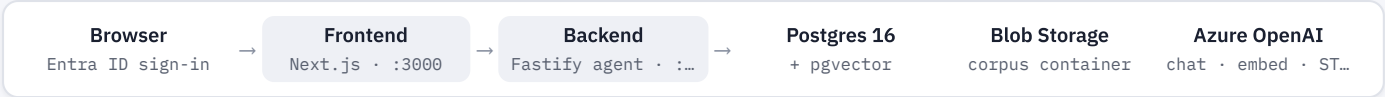


# Infrastructure Setup — One-Pager

Target Azure, single region · Profile Standard (Container Apps) · Reviewed 29 Sep 2026

What to provision to run this app for real. Grounded in the actual code, not a generic checklist — today it runs against public OpenAI with zero Azure resources provisioned.

## SHAPE OF THE SYSTEM



## PROVISION THIS

| COMPONENT                   | STATUS            | SETUP                                                                                                                                                                                                                                                                                                                            | ENV VAR                                                    |
|-----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| SSO — Entra ID              | REQUIRED          | App registration, platform <b>SPA</b> . Redirect <code>&lt;frontend&gt;/auth/microsoft/callback</code> . Scopes <code>openid profile email User.Read</code> . Client/tenant IDs are baked in at <b>build time</b> — a wrong value means a rebuild, not a config edit. Backend validates no tokens; this covers the browser only. | NEXT_PUBLIC_AZURE_CLIENT_ID<br>NEXT_PUBLIC_AZURE_TENANT_ID |
| Network / private endpoints | REQUIRED          | Private endpoint on <b>Postgres, Storage, Azure OpenAI</b> . Backend Container App joins the VNet; ingress <b>internal-only</b> — it has no auth of its own. Any gateway in front of the chat stream must disable response buffering (SSE) or streaming breaks silently.                                                         | CORS_ORIGINS                                               |
| Database                    | REQUIRED          | Postgres Flexible Server 16. Allow-list <b>vector</b> under <code>azure.extensions</code> before first boot — the app runs <code>CREATE EXTENSION</code> itself. Always <code>sslmode=require</code> .                                                                                                                           | DATABASE_URL                                               |
| Storage                     | REQUIRED          | Blob container <code>undersecretary-companion-corpus</code> (already the code default). Backend's managed identity needs <b>Storage Blob Data Contributor</b> at account scope.                                                                                                                                                  | STORAGE_BACKEND=blob<br>AZURE_STORAGE_ACCOUNT_URL          |
| AI / PTU                    | STANDARD, NOT PTU | Azure OpenAI, standard pay-as-you-go deployment per model ( <code>gpt-5</code> , <code>text-embedding-3-large</code> , TTS/STT). Skip PTU (reserved, hourly-billed capacity) — this app's traffic is bursty and departmental. Revisit only on sustained 429s or a real latency complaint.                                        | OPENAI_API_BASE                                            |
| Secrets                     | RECOMMENDED       | Key Vault for <code>OPENAI_API_KEY</code> , <code>PGPASSWORD</code> , session secrets — referenced by the Container Apps, never inlined into app settings.                                                                                                                                                                       | —                                                          |
| Compute                     | REQUIRED          | Azure Container Apps — one frontend, one backend, autoscale 1–3 replicas each. Built-in ingress is SSE-safe by default, which App Gateway is not without extra care.                                                                                                                                                             | PORT=8000                                                  |

## PROVISIONING ORDER

1. VNet + subnets → 2. Key Vault → 3. Postgres (private endpoint) → 4. Storage account + container → 5. Azure OpenAI + model deployments → 6. Entra ID app registration → 7. Backend Container App (internal ingress) → 8. Frontend Container App (public ingress) → 9. add the final redirect URI back into the app registration.

! Before reusing `deploy/` or `docs/AZURE_DEPLOYMENT_DESIGN.md` : both were inherited from a sister project ("OPANRA" / Petroleum Companion) and still carry its names — Key Vault `dof-kv-n-oap-*` , ACR repo `opanra/backend` , container `opanra-ai-petroleum-companion` , database `petroleum_companion` . Rename to this app's own resources first.